GrapeMaster: A Crop-Season-Centered Digital Platform for Traceable Vineyard Disease Management

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Abstract

Grapevine is a high-value fruit crop. In humid subtropical vineyards, epidemic diseases, particularly downy mildew, remain major constraints on yield, fruit quality, and production cost. Disease simulation, warning, diagnosis, and advisory assistance have each advanced substantially, but their outputs are often developed, validated, and adopted as separate functions, constraining their practical value for disease management. This limitation arises because vineyard disease management must respond to interactions among phenological susceptibility, infection risk, symptom diagnosis, epidemic development, fungicide application, and protection status throughout the growing season, rather than to isolated single-event outputs. To address this limitation, we developed GrapeMaster, a vineyard-level disease-management platform that uses the crop season as the core organizational unit to link daily disease pressure, plant-protection operations, and task feedback within a dynamic seasonal process. Backend verification showed that 126 crop seasons retained linked weather, phenology, risk, notification, task, fungicide-feedback, protection-state updating, advisory assistance, and message records. A representative season replay demonstrated the platform's capacity to construct a prediction–notification–task–treatment–feedback–review workflow. Module-level validation further supported the analytical readiness of downy mildew warning, broad-stage phenology display, and disease image recognition for the Guangxi context. These results indicate that GrapeMaster offers a deployable platform architecture for converting isolated digital tools into a closed-loop, traceable, and reviewable management system, with potential value for addressing complex production environments and improving the precision, standardization, and digital transformation of vineyard management.

Keywords: grapevine; downy mildew; decision support system; crop-season management; digital agriculture; advisory support; record traceability

1 Introduction

Grapevine is an important economic crop for fresh consumption, wine production, and processing. In humid subtropical vineyards, high humidity, rainfall, dew formation, and heterogeneous field microclimates readily favor epidemic diseases represented by downy mildew (Gessler et al., 2011; Yu et al., 2017, 2022; Peng et al., 2024). In severe cases, this disease can cause more than 75% yield loss (Koledenkova et al., 2022; Peng et al., 2024). However, conventional plant-protection practice often depends on fragmented field scouting, empirical spray calendars, manual symptom diagnosis, informal advisory sources with uneven reliability, and retrospective paper records. This non-integrated and weakly standardized decision process can lead to mismatches between spray timing and disease-risk periods, inappropriate pesticide concentration or mixing, and control outcomes that are difficult to trace and quantitatively evaluate (Pertot et al., 2017; Rossi et al., 2014; Oliver et al., 2023; Wise et al., 2010).

Several analytical technologies relevant to vineyard disease management have advanced substantially. Grapevine downy mildew models have progressed from empirical warning rules to mechanistic or semi-mechanistic descriptions of pathogen development and infection processes (Orlandini et al., 1993; Park et al., 1997; Caffi et al., 2011b), and warning systems can improve treatment timing when meteorological inputs, local calibration, and implementation context are adequate (Caffi et al., 2010; Caffi and Rossi, 2018; Carisse, 2016). Grapevine phenology models provide crop-stage information for interpreting host susceptibility, warning windows, canopy development, and operation timing (Caffarra and Eccel, 2010; Parker et al., 2011; García de Cortázar-Atauri et al., 2017; Wang et al., 2020). Image-recognition and advisory systems can convert visible symptoms into structured diagnostic evidence and management-oriented explanation (Yan et al., 2024, 2025; Shafay et al., 2026; Rai et al., 2026). These technologies have substantial value in different settings. The challenge for extension and industrial deployment is how to coordinate their separate outputs within a shared seasonal management context.

More broadly, digital agriculture and decision support system (DSS) implementation studies increasingly show that production deployment is not determined by model performance alone. Deployed agricultural DSS platforms have begun to combine monitoring, analytical models, advisory delivery, and user-facing workflows: vite.net addressed the implementation problem in vineyard DSSs through integrated vineyard monitoring, model-based decision support, and two-way communication (Rossi et al., 2014); MISFITS-DSS harmonized regional grapevine downy mildew risk assessment for public plant-protection services (Bregaglio et al., 2022); and Fuxi Brain illustrated a whole-cycle intelligent management platform based on IoT perception, multimodal fusion, and generative decision making in maize production (Chen et al., 2026). Together, these examples indicate that DSSs intended for production use must move beyond model output alone and require stable data ingestion, modular architecture, reliable service operation, user-facing task processes, feedback mechanisms, record tracking, workflow evaluation, and traceability. However, for perennial fruit-crop production contexts, they also expose a platform-level architectural problem: how analytical outputs can be organized into a crop-season-centered workflow so that disease-management decisions can be delivered, recorded, executed, updated, and reviewed within a traceable seasonal frame.

Therefore, for grape production, the unresolved problem is not only whether individual models or diagnostic tools can perform accurately, but whether their outputs can be organized into a verifiable seasonal decision chain. Otherwise, after a warning is generated, the system may not be able to trace whether the warning was delivered, reviewed, converted into an operation, reflected in subsequent protection status, and used as the updated management context for subsequent risk assessment and evaluation.

To address this gap, we developed and deployed GrapeMaster, a closed-loop and traceable vineyard disease-management platform that uses the crop season as the core organizational unit. Here, the crop season denotes the annual production cycle from budburst to fruit maturity and harvest, and is used as a shared unit for storing vineyard and crop context and linking crop development, risk interpretation, warning delivery, task execution and feedback, diagnostic and advisory records, and message storage. The central research question was whether this platform-implemented architecture can convert independent analytical and modular functions into a closed-loop and verifiable workflow linking prediction, action, feedback, symptom or advisory evidence, and review across the crop season. We evaluated this question through three verification levels: the crop-season data model and database relationships that implement this management unit, workflow constraints and state-transition logic, and backend-record verification using linkage checks and representative season replay. GrapeMaster provides a platform-level pathway and methodological basis for consistent, standardized, traceable, and evaluable disease management in perennial fruit production.

2 Materials and methods

2.1 Study design, verification evidence, and vineyard context

2.1.1 Study design and verification strategy

This study describes the design, deployment, and record-based evaluation of GrapeMaster, a crop-season-centered digital platform for grapevine disease management. The platform workflow includes crop-season setup, weather and phenology updating, disease-risk warning, plant-protection task management, post-treatment protection-state updating, image-assisted disease diagnosis, large language model (LLM)-assisted consultation, and management-record archiving.

The evaluation used two complementary verification streams. Backend records from the deployed platform were used as record-level evidence to evaluate whether mobile functions, backend services, database objects, scheduled workflows, notification and task services, image-upload pathways, advisory records, and crop-season records were linked as a functioning management workflow. Module-level validation evidence was used to characterize analytical components that can operate within this workflow.

The analytical layer includes a downy mildew initial-infection model for first seasonal infection-period warning, a phenology module for crop-stage interpretation, a grape-focused image-recognition module for symptom evidence, and an LLM advisory module for season-long consultation and management-oriented explanation. These modules were evaluated according to their platform roles and input–output compatibility with the crop-season workflow; additional module-level technical details are provided in Supplementary Table S1.

2.1.2 Study area and field data sources

The analytical modules were configured and evaluated for humid subtropical vineyard management using field data from Guangxi, China. This regional data set included meteorological records, grapevine phenological observations, airborne sporangia monitoring, symptom observations, and management records collected from field experiments and related vineyard observations. These regional data supported the downy mildew risk engine based on the first seasonal infection model for subtropical vineyards (FSIM-S), phenology calibration, and grape disease image expansion used in the platform-compatible analytical configuration. Field and experimental data were analyzed separately from the backend operational records generated by the deployed GrapeMaster platform. Consequently, module-level results are interpreted for the Guangxi humid subtropical vineyard configuration.

2.2 Platform architecture and crop-season data model

2.2.1 Software architecture and service layers

GrapeMaster uses a layered architecture in which the mobile client, backend services, analytical model services, persistent storage, and external data sources are separated into logical service layers. The crop season is the central platform design unit linking a vineyard field with cultivar information, cultivation method, vine age, weather history, phenology outputs, disease-risk simulations, fungicide applications, plant-protection tasks, notes, image records, messages, and notifications. The platform method therefore consisted of defining the crop-season object, implementing database links and workflow transitions around this object, and verifying whether exported backend records preserved the expected identifiers and state transitions. At the implementation level, the formal crop-season object, database relationship rules, workflow state-transition rules, and verification rules are specified in Supplementary Tables S2–S5.

Table 1 summarizes the five logical layers used for application interaction, data ingestion, analytical modelling, image-based diagnosis, alerting, and operational feedback. Detailed software dependencies and implementation components are listed in Supplementary Table S6.

Table 1. Logical architecture layers of GrapeMaster used in the current study.

Layer	Main components	Main responsibilities
Application and presentation layer	Mobile modules for field, crop-season, task, recognition, map, note, authentication, weather, and notification functions	Field input, weather and warning display, plant-protection task operation, image upload, consultation entry, and grower-facing interaction
Data ingestion and storage layer	Backend API services, authentication, persistent storage, scheduled jobs, master-data tables, and business objects	API routing, authentication, data persistence, scheduled updates, field records, crop-season records, and operational records
Weather, phenology, and disease analytics layer	Weather preprocessing, phenology workflow, spraying-suitability workflow, and disease-model interface	Weather transformation, crop-stage simulation, disease-risk request construction, treatment-window generation, and risk interpretation
Image recognition and LLM advisory layer	Image upload pathway, visual disease-recognition service, consultation interface, and advisory-response records	Symptom-image processing, disease-category output, visual evidence storage, LLM-assisted explanation, and season-long advisory support
Task recommendation, alert, and feedback layer	Notification services, task services, fungicide and pesticide records, treatment windows, and protection-state feedback	Warning delivery, task creation, task completion, overdue handling, fungicide-history conversion, and risk recalculation after completed operations

2.2.2 Field and crop-season records

The data model was organized around two primary agricultural entities: the vineyard field and the crop season. The field record represents the persistent spatial and ownership unit, including field identity, area, boundary, location, administrative attributes, sharing state, and management status. Detailed field-record attributes and related crop-season data-model attributes are listed in Supplementary Table S7.

The crop-season record links the field with grape cultivar, cultivation method, vine age, field coordinates, field size, and cultivar metadata. At the implementation level, it connects weather records, phenology outputs, disease-risk outputs, fungicide and pesticide records, plant-protection tasks, irrigation tasks, notes, notifications, image records, and user-facing summaries under the crop-season management unit. This structure reflects the distinction between persistent field identity and season-specific disease risk, weather, phenology, protection history, observations, and management actions.

For one crop season c, the implementation-level traceability representation was formalized as:

$$T_c = (F_c, M_c, W_c, P_c, R_c, N_c, O_c, B_c, E_c, A_c), \tag{1}$$

where F_c is the linked field record, M_c is crop-season metadata, W_c is weather input, P_c is phenology state, R_c is disease-risk and request-response output, N_c is notification or event delivery, O_c is task or field-operation state, B_c is fungicide or pesticide feedback, E_c is symptom or image evidence, and A_c is advisory or message evidence. The minimum traceability chain evaluated in this study was:

$$F_c \rightarrow M_c \rightarrow (W_c, P_c) \rightarrow R_c \rightarrow N_c \rightarrow O_c \rightarrow B_c, \tag{2}$$

with E_c and A_c treated as additional symptom or advisory records when present. A crop season was considered analytically linked when W_c, P_c, and R_c shared the same crop-season key; risk-to-operation linkage was present when N_c or O_c could be retrieved under the same retained user, field, or crop-season context; and treatment-feedback linkage was present when B_c was connected to O_c through task identifiers.

Figure 1 shows the crop-season-centered traceability structure implemented in GrapeMaster. The main prediction–action–feedback–replay chain and the optional symptom-image and advisory-record branches are organized around the same crop-season management unit.

Master-data tables were used as controlled reference tables for field and crop-season records rather than as independent management objects. Grape variety, cultivar susceptibility, cultivation method, BBCH stage, disease type, fungicide, pesticide, and reminder-rule tables provide standardized attributes when field, crop-season, task, risk, or advisory records are created. The implementation preserves both structured business records and model payloads so that weather, phenology, disease-risk, and request-response records remain available for display, recalculation, and audit.

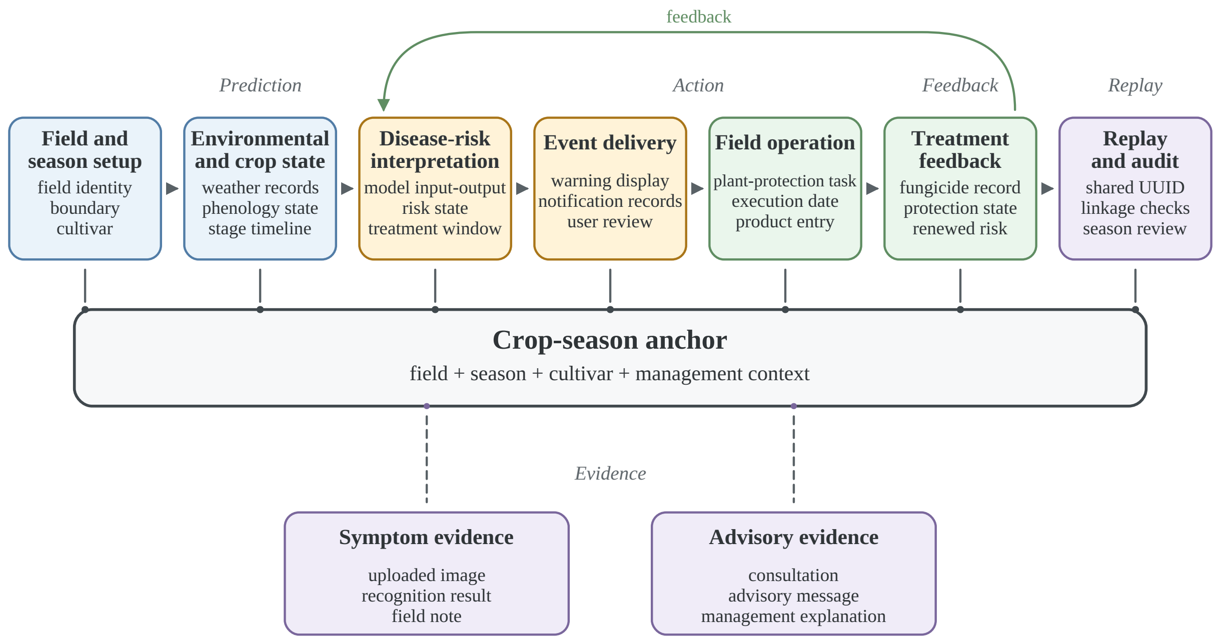


Figure 1. Crop-season-centered traceability architecture implemented in GrapeMaster. The crop season serves as the platform management unit linking field setup, environmental and crop state, disease-risk interpretation, event delivery, field operation, treatment feedback, optional symptom and advisory evidence, and replay/review records. The feedback path indicates that completed plant-protection records can update later protection-state or risk interpretation, while replay/review records support reconstruction of the prediction–action–feedback chain.

2.3 Weather ingestion and crop-season preprocessing

For each active crop season, scheduled workflows retrieve historical and forecast weather using the field location. In the current implementation, weather data were obtained from the Open-Meteo weather API (https://open-meteo.com/) using field coordinates. The weather workflow combines historical observations with forecast records where available. Hourly and daily variables include temperature, humidity, dew-point or wetness-related variables, precipitation, wind conditions, and daily summary variables. Missing values are handled before analytical use so that downstream workflows receive complete date-aligned input series.

The backend transforms weather data into three working schemas. Daily mean temperature is converted into phenology-model input; hourly weather is converted into the disease-model schema, including datetime, dew point, precipitation, relative humidity, air temperature, and wind speed; and short-term hourly records are used to evaluate spraying suitability. This preprocessing belongs to the GrapeMaster platform workflow and defines the input schema required by the platform-compatible analytical modules.

2.4 Platform-compatible analytical modules and module-level evaluation

2.4.1 Phenology simulation and Guangxi calibration

The phenology workflow estimates grapevine growth-stage progression from daily temperature and crop-season metadata. The backend calculates daily thermal accumulation and maps accumulated development to broad growth-stage classes related to the BBCH (Biologische Bundesanstalt, Bundessortenamt and Chemical industry) scale. The output provides a crop-development timeline for mobile display and a daily crop-stage sequence for disease-risk interpretation, warning windows, and management-task scheduling.

For regional evaluation, Guangxi field phenology observations were mapped to broad major-stage classes: early season or budburst, leaf development, inflorescence emergence, flowering, berry development, ripening, and post-harvest or senescence. Daily mean temperature was converted to cumulative growing degree days (GDD) after a meteorological biofix:

$$\mathrm{GDD}_t = \sum_{d=1}^{t} f_{\mathrm{WE}}\left(T_d\right), \tag{3}$$

where T_d is daily mean temperature and $f_{\mathrm{WE}}(\cdot)$ is the Wang–Engel temperature-response function. The accumulated value was mapped to the broad-stage sequence using cultivar-specific major-stage thresholds. For mobile display, the predicted major stage was supplemented with within-stage progress and a transition label when the thermal state was close to an adjacent stage boundary.

The Guangxi observations were evaluated using grouped validation by site-year and cultivar-stratified grouping where possible. The primary metrics were exact major-stage agreement, user-visible agreement after transition-window display, adjacent-stage agreement, severe misclassification rate, and mean absolute stage distance. The cultivar wk was represented by one sparse site-year and was reported separately.

2.4.2 FSIM-S downy mildew first-infection risk engine

The disease-risk service receives weather, crop-stage sequence, cultivar susceptibility, target disease information, and fungicide-history inputs through the platform interface. It returns date-aligned disease-risk states, field-risk states, protection states, recommendations, and treatment windows.

For GrapeMaster, the disease-risk module was configured with FSIM-S for humid subtropical grape-growing areas in Guangxi. The module predicts the first seasonal infection of grapevine downy mildew under conditions where early-season infection may be driven by recurrent airborne asexual sporangia rather than by a discrete oospore-driven primary infection event. It was calibrated and validated using 2022–2024 site-year observations from Guangxi, including phenology, airborne sporangia, symptom onset, management records, and meteorological data.

FSIM-S uses a mechanistic infection chain adapted from Rossi primary-infection modelling and related sporangia-based infection modelling work (Rossi et al., 2008; Caffi et al., 2009, 2011a; Brischetto et al., 2021). In the subtropical setting, airborne sporangia are treated as the relevant inoculum source for the first seasonal infection, and the model represents the main steps from sporangia availability and deposition to infection establishment, incubation, and symptom expression. The FSIM-S configuration evaluated in this study uses field-derived spore pressure to regulate the infection-triggering threshold. A daily sporangia pressure field $S(d)$ is constructed from field spore monitoring data after interpolation, normalization, and smoothing. Hourly pressure is represented as a three-day rolling mean:

$$P_t = \frac{1}{3} \sum_{k=0}^{2} S\left(d_t - k\right), \tag{4}$$

where P_t is the spore-pressure index at hour t, and d_t is the corresponding calendar day.

Infection is triggered when the wetness-duration term and its temperature response exceed a spore-pressure-adjusted effective threshold:

$$\begin{aligned} \mathrm{IC}_t^* &= \begin{cases} \mathrm{IC}\exp\left(-\alpha_{\mathrm{pos}} P_t\right), & P_t \geq 0, \\ \mathrm{IC}\exp\left[\alpha_{\mathrm{neg}}\left(-P_t\right)\right], & P_t < 0, \end{cases} \\ \mathrm{infection}_t &= \mathbb{I}\left(\mathrm{WD}_t\,\mathrm{TWD}_t \geq \mathrm{IC}_t^*\right), \end{aligned} \tag{5}$$

where WD_t is the wetness-duration component, TWD_t is the temperature response during the wetness period, IC is the base infection threshold, α_{pos} and α_{neg} are calibration parameters for positive and negative spore-pressure effects, and $\mathbb{I}(\cdot)$ is an indicator function. Positive spore pressure lowers the effective infection threshold, whereas low pressure raises it.

The disease-risk workflow was limited to downy mildew because FSIM-S was calibrated for this target disease. Other grape diseases were represented in image-recognition or advisory functions but were not disease-risk prediction targets in this study. Weather inputs can be supplied through public weather services, grid weather products, regional weather data, local weather stations, or other sources that can be transformed into the platform schema. The spore-pressure term represents an FSIM-S model-level enhancement and an optional sensor-enhanced pathway for vineyards that deploy spore-trapping equipment; it was not required as a live input for routine platform operation. Supplementary Figure S1 presents the platform-level inputs, transformations, and user-facing outputs required for integrating the disease-risk engine with the GrapeMaster workflow.

2.4.3 Grape disease recognition and LLM-assisted advisory

The image-based workflow supports symptom-level evidence after disease signs are observed or suspected. Users can upload field images through the mobile client, and the image-recognition service returns disease or healthy categories with associated diagnostic or management text for display. Image outputs can be linked to crop-season notes, disease reports, and advisory records, allowing visual evidence to remain within the same crop-season context as other management records.

The image-recognition and advisory module follows the published CDIP-ChatGLM3 dual-model framework, which combines a deep-learning image classifier with a fine-tuned ChatGLM3-based advisory model for crop disease identification and management guidance (Yan et al., 2025). In this study, the framework was adapted to the grape scenario by expanding the grape disease image dataset with additional field photographs and two grape disease categories. The platform-compatible visual model was selected according to grape-focused validation performance.

For image recognition, each candidate model outputs class probabilities using a softmax layer and is trained with cross-entropy loss:

$$\begin{aligned} p\left(y=k \mid x\right) &= \frac{\exp\left(z_k\right)}{\sum_{j=1}^{K}\exp\left(z_j\right)}, \\ \mathcal{L}_{\mathrm{CE}} &= -\sum_{k=1}^{K} y_k \log p\left(y=k \mid x\right), \end{aligned} \tag{6}$$

where x is the input image, z_k is the logit for disease class k, K is the number of grape disease or healthy categories, and y_k is the one-hot class label. Model selection is based on grape-focused validation metrics, including accuracy, precision, recall, and F1-score:

$$m^* = \arg\max_{m \in \mathcal{M}} \mathrm{Score}_{\mathrm{val}}\left(m\right), \tag{7}$$

where $\mathcal{M}$ is the set of candidate vision models and $\mathrm{Score}_{\mathrm{val}}$ is the selected validation criterion, such as accuracy or macro-F1. The selected classifier provides the disease label and confidence score used by the advisory interface.

The LLM-assisted advisory component follows the CDIP-ChatGLM3 interaction structure: image-recognition outputs are converted into disease-specific consultation context, and the fine-tuned language model generates management-oriented advisory text. In the platform workflow, LLM messages and advisory outputs are stored or linked to the crop-season context where possible. Supplementary Figure S2 summarizes the image-recognition and advisory workflow.

2.5 Module provenance and claim boundaries

Table 2. Provenance, platform roles, and evidence-to-claim boundaries of the main GrapeMaster components. Backend records and module-level validation evidence are reported separately because they support different claim levels.

Module	Source or provenance	Platform role	Evidence type and claim boundary
Platform workflow	Deployed GrapeMaster backend, mobile client, database, scheduled jobs, and notification/task services	Links field, crop season, weather, phenology, risk, task, fungicide, image, advisory, and record objects	Backend-record evidence for workflow operability and data linkage; not evidence of production effects
Phenology module	Weather-driven phenology workflow calibrated with Guangxi field observations	Crop-stage timing for risk interpretation, advisory timing, and task scheduling	Offline module-level evidence for platform-compatible broad-stage crop context
FSIM-S risk engine	Team-developed downy mildew first-infection model calibrated and validated with Guangxi field data	First seasonal infection-period warning for task triggering and protection-state interpretation	Offline module-level evidence for a platform-compatible downy mildew risk component
Grape disease recognition	Platform-compatible grape disease visual model based on the published CDIP-ChatGLM3 framework	Post-symptomatic image evidence and disease category output	Offline validation evidence for a platform-compatible grape disease image component
LLM advisory	Fine-tuned ChatGLM3-based advisory model from the CDIP-ChatGLM3 framework	Consultation, explanation, and agronomic advisory text	Knowledge-support evidence for grape disease consultation cases; does not replace expert or regulatory decisions

Table 3. Evidence-to-claim boundaries used in the GrapeMaster evaluation.

Verification source	Supported claim	Boundary or excluded claim
Deployed backend records	GrapeMaster generated linked crop-season, weather, phenology, risk, notification, task, fungicide, image, advisory, and record objects in the operational database	Does not prove post-replacement module performance, user adoption, pesticide reduction, disease-control efficacy, or economic benefit
Offline module-level validation	The FSIM-S, broad-stage phenology, and grape disease image-recognition components have platform-compatible inputs, outputs, and module-level analytical evidence for the Guangxi context	Does not represent full-season production validation after replacing or updating modules in the deployed platform
Representative crop-season replay	A selected crop-season management unit can retrieve weather, phenology, disease-risk, notification, task, and fungicide records as a traceable risk-to-feedback chain	Does not represent population-level behaviour or a controlled agronomic outcome test

2.6 Risk-task-protection workflow and evaluation boundaries

2.6.1 Risk-task-protection feedback logic

The risk-task-protection workflow starts from the daily disease-risk output generated for an active crop season. The platform translates model output into user-facing risk states, such as protected, low-risk, favourable, or high-risk periods. When weather, phenology, and disease-risk outputs indicate a favourable infection window, the crop-season page and notification interface display the corresponding risk state and prompt the user to review the field condition and treatment window. The notification is linked to the crop season rather than to a stand-alone warning message. From the warning page, the user can enter the plant-protection workflow, inspect the suggested treatment timing, and create a plant-protection task if intervention is required. This design connects risk viewing, decision review, and task creation within the same crop-season context.

After a plant-protection task is created, the user can keep it pending, complete it after field operation, or leave it unexecuted. A completed task records the execution date and pesticide or fungicide information and is then used by the platform as management feedback. This rule follows the practical plant-protection assumption that fungicide applications can provide a temporary protection window whose duration depends on product properties, application timing, host growth, rainfall, and disease pressure (Caffi and Rossi, 2018; Bleyer et al., 2020; Oliver et al., 2023; Warneke et al., 2020). In this workflow, a completed fungicide-related task activates a rule-based protection period or low-risk management state. During this period, the crop-season risk display is updated to a protected or low-risk state.

When the protection period ends, the platform resumes dynamic risk prediction using the updated weather, phenology, disease-risk input, and management history. If a new favourable or high-risk window appears, the system again displays the risk state and prompts the user to review the need for a new plant-protection task. The resulting cycle is: risk prediction, risk-state display, user notification, task creation, task completion, protected-state update, and renewed risk prediction.

The image-recognition and LLM-assisted advisory modules provide an additional entry point when users observe symptoms during the crop season. A user can upload a field image for disease identification, and the platform returns a grape-disease diagnosis result that can be reviewed together with the current crop-season risk state. For downy mildew, this image-based confirmation can be aligned with the risk-prediction workflow. For other supported grape diseases, such as powdery mildew, the module provides diagnostic support without being treated as an equivalent risk-forecasting workflow.

The LLM-assisted function is positioned as a supplementary advisory and knowledge-support channel. Within the crop-season context, it supports agronomic consultation, interpretation of recognition results, and communication of standard management guidance. Risk forecasting and image-based disease diagnosis remain handled by the dedicated analytical modules; LLM outputs are treated as auxiliary information for learning, communication, and management planning rather than as a replacement for validated models, expert diagnosis, pesticide labels, or local regulations.

2.6.2 Backend records, crop-season replay, and evaluation boundaries

Backend database exports from GrapeMaster were used to assess whether the platform had generated the record types required by the workflow. The exported tables included user and profile tables, field records, crop-season records, notifications, plant-protection tasks, irrigation tasks, fungicide and pesticide records, weather-related records, phenology records, disease-risk records, image-recognition and advisory records, disease-report records, note records, and message records. These tables were treated as operational records for evaluating workflow operability and traceability. The export represented the operational records available at the time of data extraction and was used as a cross-sectional record audit rather than as a longitudinal adoption or production-outcome analysis.

Account-level screening was applied before summarizing GrapeMaster records. Accounts without business records or without both field and crop-season records were excluded because they could not support a crop-season-centered workflow. One non-target-region account was also removed from the vineyard evaluation set. Detailed screening rules are listed in Supplementary Table S8. After screening, 47 accounts remained from 95 deduplicated accounts and were used as the candidate account set for describing GrapeMaster operational-record coverage.

The screened records were used for four workflow checks: field and crop-season creation and linkage; attachment of weather, phenology, risk, and notification records to crop seasons; creation and status updating of plant-protection tasks and pesticide/fungicide records; and linkage of image, advisory, disease-report, and note records to management contexts.

The representative replay anchor was selected as a chain-complete audit case from retained crop seasons that had the minimum traceability chain required for risk-to-feedback reconstruction: linked weather, phenology, disease-risk or request-response payloads, notification records, completed plant-protection tasks, and fungicide-product records. The replay was not selected to estimate population-level user behaviour, task-completion frequency, agronomic efficacy, pesticide reduction, or adoption rate. It was used to test whether the expected workflow objects could be reconstructed through shared crop-season or task identifiers and whether optional symptom or incidence records were present or absent under the selected anchor.

The study uses separate evidence sources for platform workflow records and analytical modules. The screened backend records and workflow outputs were used as record-level verification evidence for field creation, crop-season management, weather and phenology records, disease-risk output, notification delivery, plant-protection task handling, fungicide-feedback logic, image upload, LLM consultation, and management-record storage. Module-level validation evidence was evaluated separately for the downy mildew initial-infection model, phenology module, grape disease image-recognition module, and LLM advisory module. The resulting evidence therefore supports deployed workflow operability, record traceability, representative crop-season replay, and module-level analytical readiness. It does not evaluate post-replacement full-season production performance, user adoption, pesticide reduction, disease-control efficacy, or economic outcomes.

3 Results

3.1 Backend-record verification of crop-season-centered linkage

The cleaned GrapeMaster backend export contained 95 deduplicated registered accounts. Among them, 47 accounts contained both field records and crop-season records; the remaining 48 accounts did not contain field-crop-season linkage data.

Within the 47 accounts with field-crop-season linkage, the backend contained 139 vineyard fields and 128 crop seasons. Analytical-state records were linked at the crop-season level, including 126 weather payloads, 126 phenology payloads, and 252 disease-risk or model request-response payloads. Event and operation records were also retained, including 3590 notifications, 151 plant-protection or irrigation tasks, and 234 fungicide or pesticide records.

The optional symptom and advisory branches contained 94 field-note or disease-report records, 2808 image-analysis records, 1360 advisory or message records, and 27 forum records. Table 4 reports the record-level linkage evidence and claim boundaries across the retained backend record classes; the backend linkage topology is provided in Supplementary Figure S4.

Among the retained backend records, 124 field records from the 47 retained accounts had valid coordinates and crop-season linkage and were mapped as 27 regional aggregates (Figure 2). The teal bubbles in Figure 2 show these mappable platform field records, with bubble size indicating the number of field records in each regional aggregate. Red points indicate independent experimental sites used for module-level validation and were not part of the backend-record denominator.

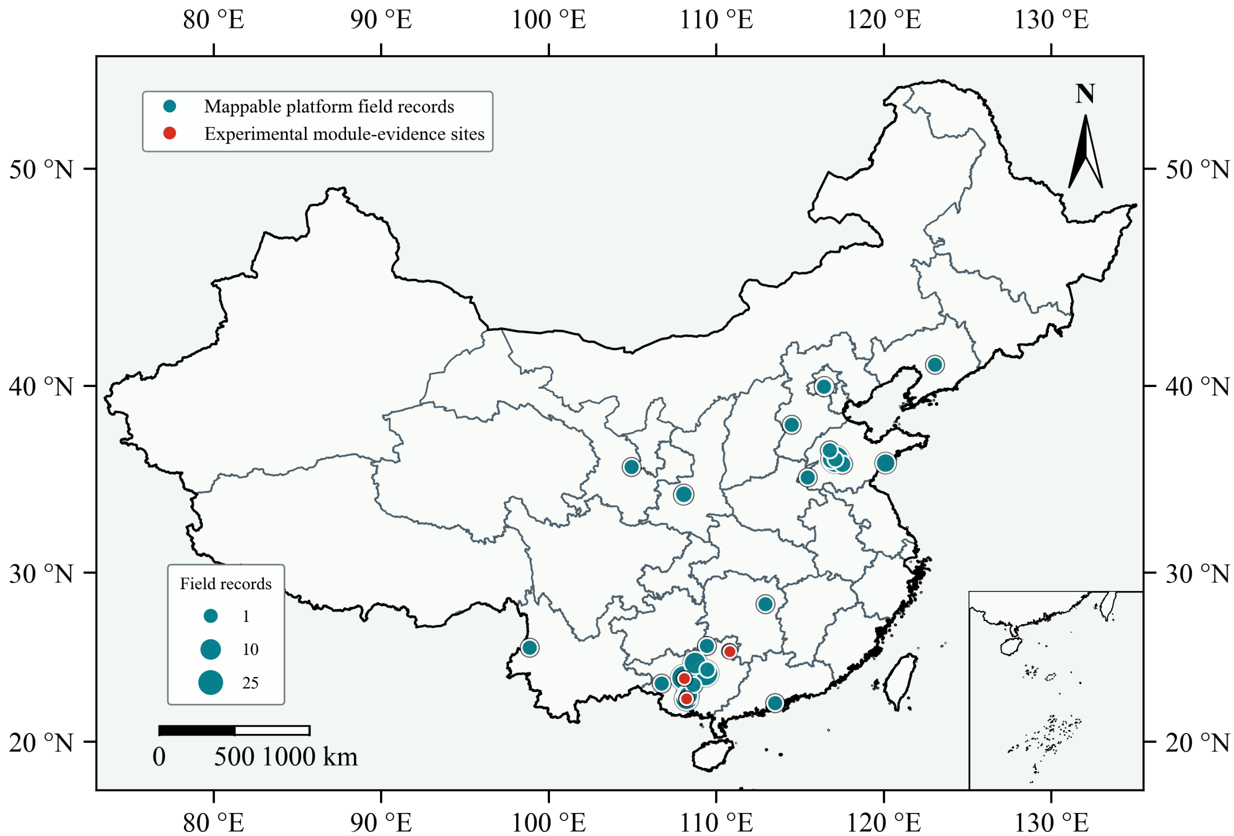


Figure 2. Spatial context of mappable platform field records and independent experimental module-validation sites. Teal bubbles show 124 mappable field records from the retained backend export aggregated into 27 regions, with bubble size indicating the number of records per region. Red points indicate experimental sites used for module-level validation and are not part of the backend-record denominator.

Table 4. Backend-record verification of crop-season-centered linkage in the retained GrapeMaster export.

Verification level	Retained records	Linkage basis	Result supported
Screened workflow set	47 of 95 deduplicated accounts retained after screening	At least one field and one crop season	Accounts available for crop-season workflow evaluation
Field-season denominator	139 fields and 128 crop seasons	Retained account, field UUID, and crop-season UUID	Field and crop-season records established as the verification base
Crop-season state records	126 crop seasons with linked weather, phenology, risk, and request-response payloads	Shared crop-season UUID	Environmental, crop-stage, and disease-risk states reconstructable by crop season
Event and operation records	3590 notifications and 151 plant-protection or irrigation tasks	Retained user, field, or crop-season context	Event-delivery and field-operation records retained
Treatment-feedback records	234 fungicide or pesticide records linked to 141 plant-protection tasks	Task UUID	Product records connected to completed or recorded task contexts
Symptom and advisory records	94 field-note or disease-report records, 2808 image-analysis records, 1360 advisory or message records, and 27 forum records	Account, field, crop-season, image, or message identifiers	Optional symptom/advisory branches retained; completeness varies by season
Claim boundary	No controlled production comparison in the backend export	Not applicable	No efficacy, pesticide-reduction, adoption, or economic outcome inferred

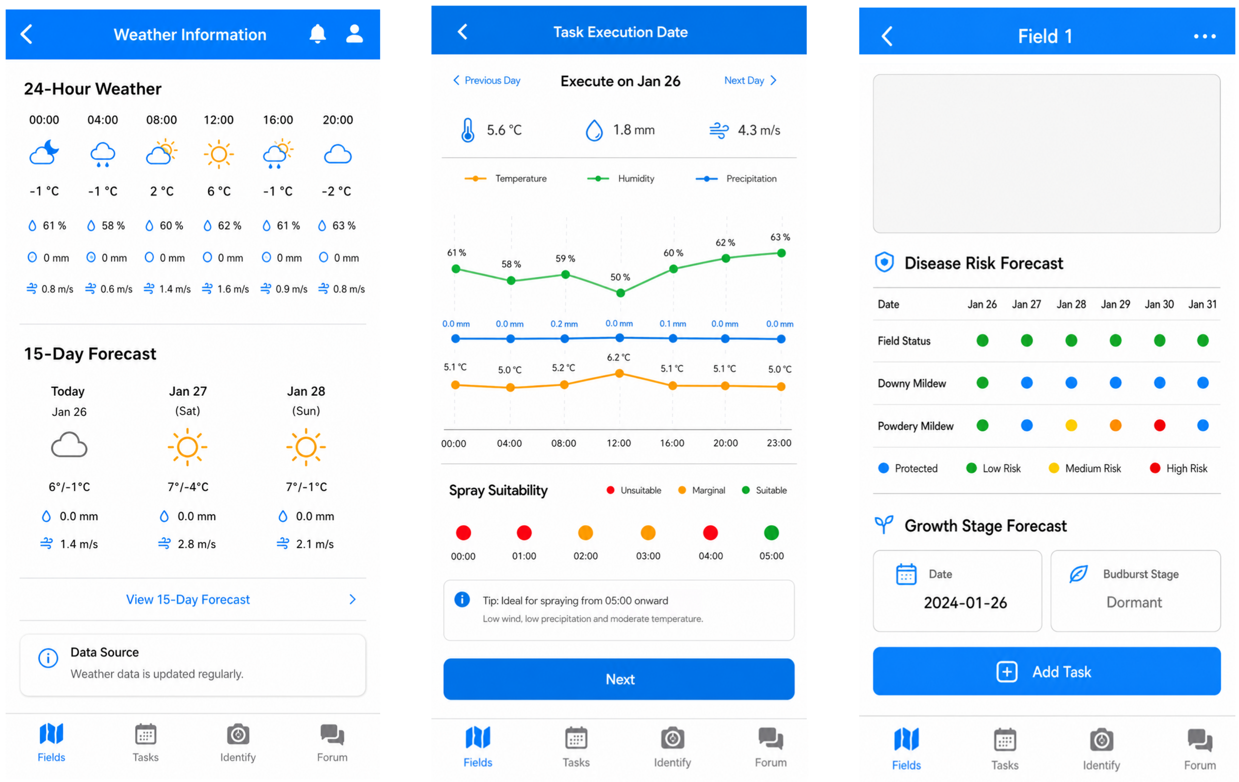
UUID, universally unique identifier.

3.2 Crop-season workspace entry and state display

The mobile client displayed vineyard fields and crop seasons as the entry points for crop-season management functions. Field-management pages provided field creation, inspection, selection, sharing, deletion, recovery, crop-season access, and links to weather, phenology, disease-risk warning, notifications, tasks, notes, and image-related records (Supplementary Figure S3).

Within this workspace, weather, operation-timing, growth-stage, and summary field-risk outputs were displayed under the selected crop-season context. The weather page presented near-term field-specific weather information (Figure 3a). The operation-timing page displayed hourly weather conditions and spray-suitability information for field work planning (Figure 3b). The crop-season workspace summarized field-risk and growth-stage states under the selected field context (Figure 3c).

The weather and phenology interfaces corresponded to the linked weather and phenology payloads reported in Table 4; disease-risk delivery, task feedback, image recognition, and advisory interaction records are reported in Section 3.3.



(a) (b) (c)

Figure 3. Crop-season workspace for weather and phenology state in GrapeMaster. (a) The weather page displays near-term field-specific weather information. (b) The operation-timing page displays hourly weather and spray-suitability information. (c) The crop-season workspace displays field-risk and growth-stage summaries under the selected field context.

3.3 Risk delivery, operation feedback, and record capture

Disease-risk outputs were displayed as a date-aligned mobile timeline for crop-season review. The timeline presented downy mildew risk together with field-level risk states in the Mingyang crop-season case (Figure 4a), in a Guangxi crop-season case without fungicide-feedback records (Figure 4b), and in the same case showing the protected-state sequence associated with fungicide-feedback records (Figure 4c).

The warning and notification interfaces displayed crop-season and disease-risk events as reviewable notification records. The disease-alert dialog displayed a disease-risk event (Figure 5a), the management-notice dialog displayed a task-related prompt (Figure 5b), and the notification list retained event entries for later review (Figure 5c). These notification records corresponded to the retained event-delivery records summarized in Table 4.

Task interfaces recorded field-operation information after crop-season events were reviewed. The task list displayed field-operation records and status (Figure 6a), the product-selection page retained fungicide or pesticide entries linked to plant-protection tasks (Figure 6b), and the in-tank mixing page recorded sprayer capacity, product dosage, plot area, water volume, and planned date (Figure 6c).

The symptom-image and advisory interfaces displayed symptom-image upload (Figure 7a), disease-recognition output (Figure 7b), and advisory interaction functions (Figure 7c). These records corresponded to the image-analysis and advisory/message branches reported in Table 4.

Across these interfaces, the displayed record sequence followed the configured workflow order from risk-state display to event delivery, task recording, treatment-feedback capture, and optional symptom-image or advisory-record entry.

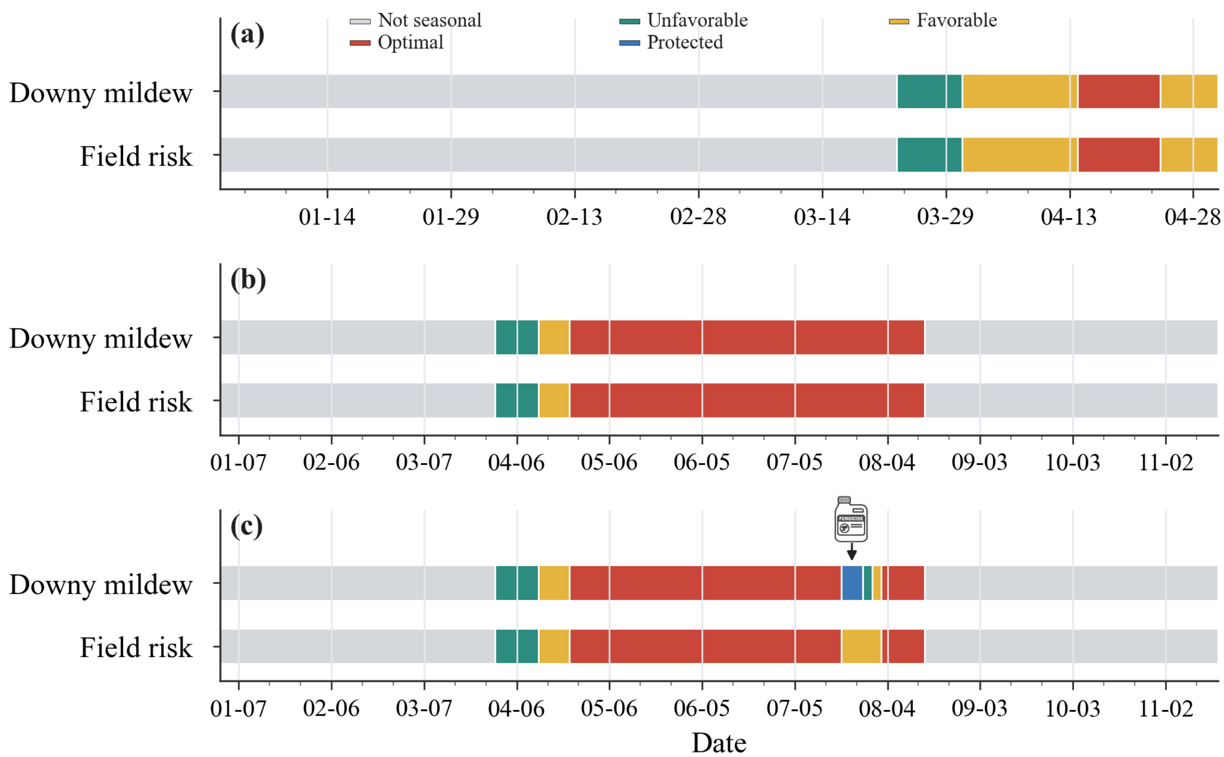
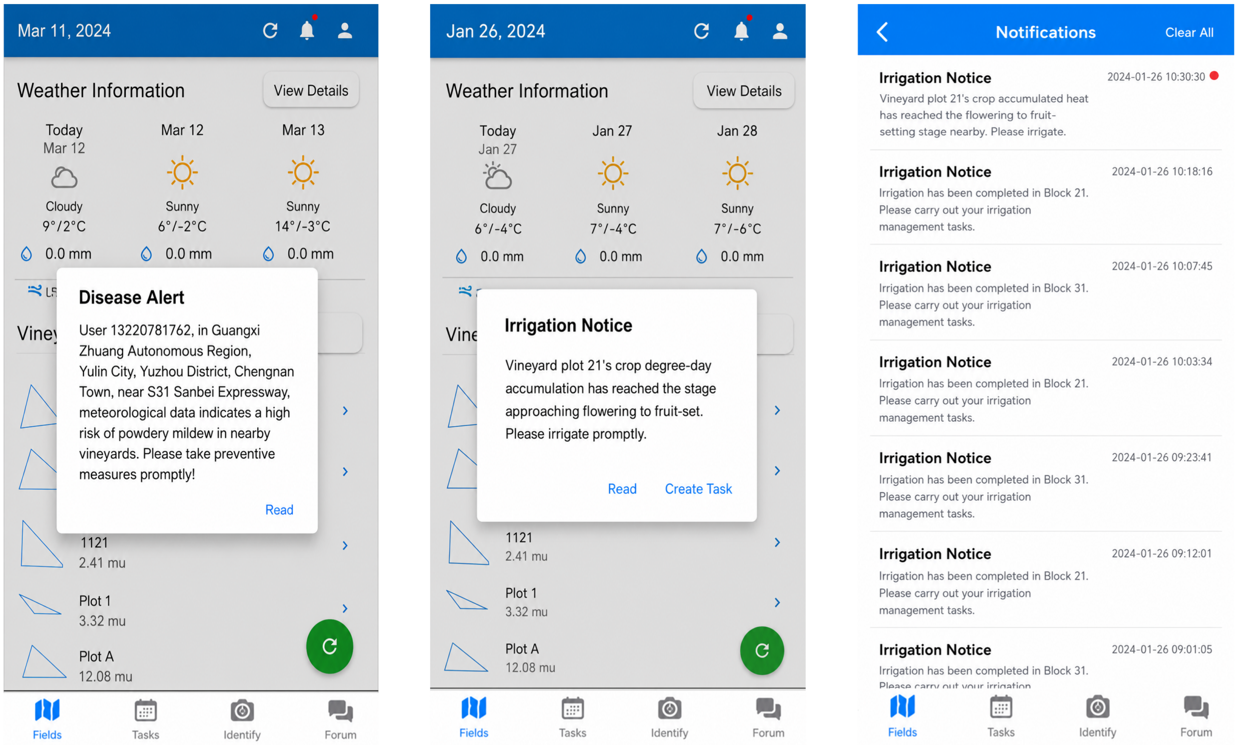
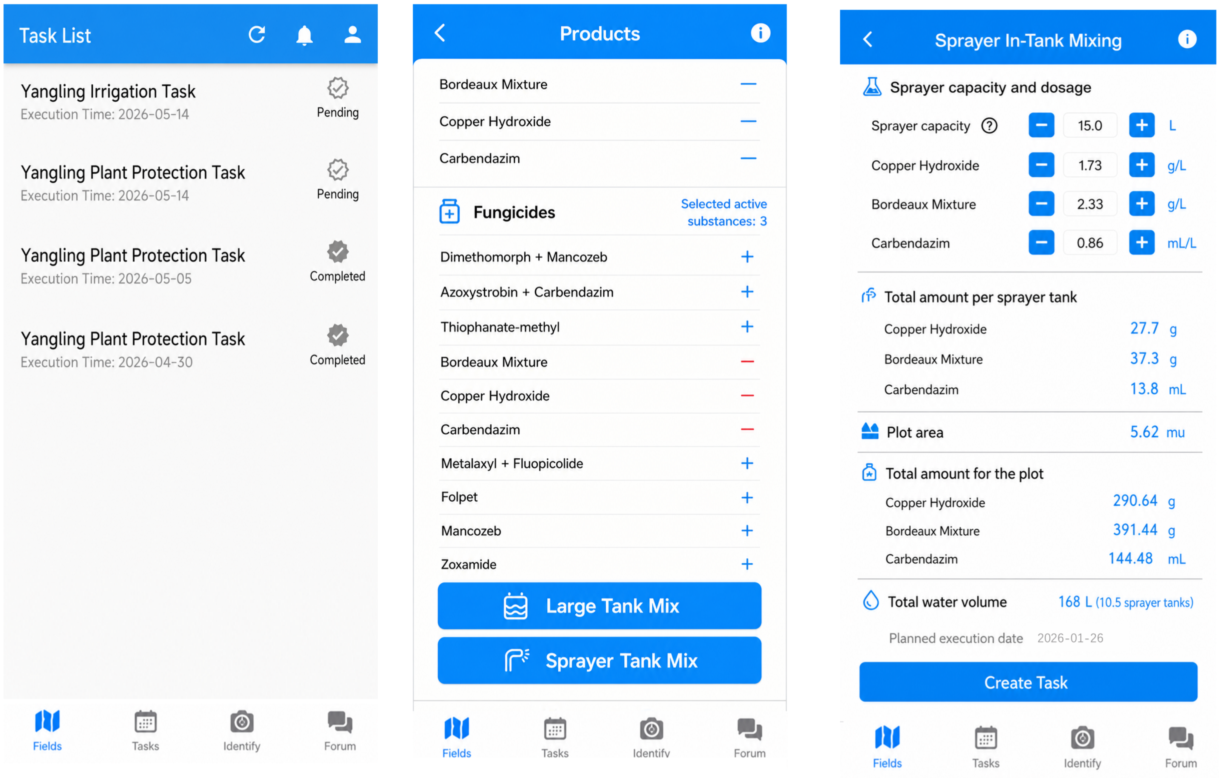


Figure 4. Disease-risk timeline in the GrapeMaster mobile workflow. (a) Mingyang 2024 shows date-aligned downy mildew and field-risk states. (b) A Guangxi crop-season case without fungicide-feedback records shows the untreated risk-state sequence. (c) The same Guangxi crop-season case with fungicide-feedback records shows the corresponding protected-state updates.



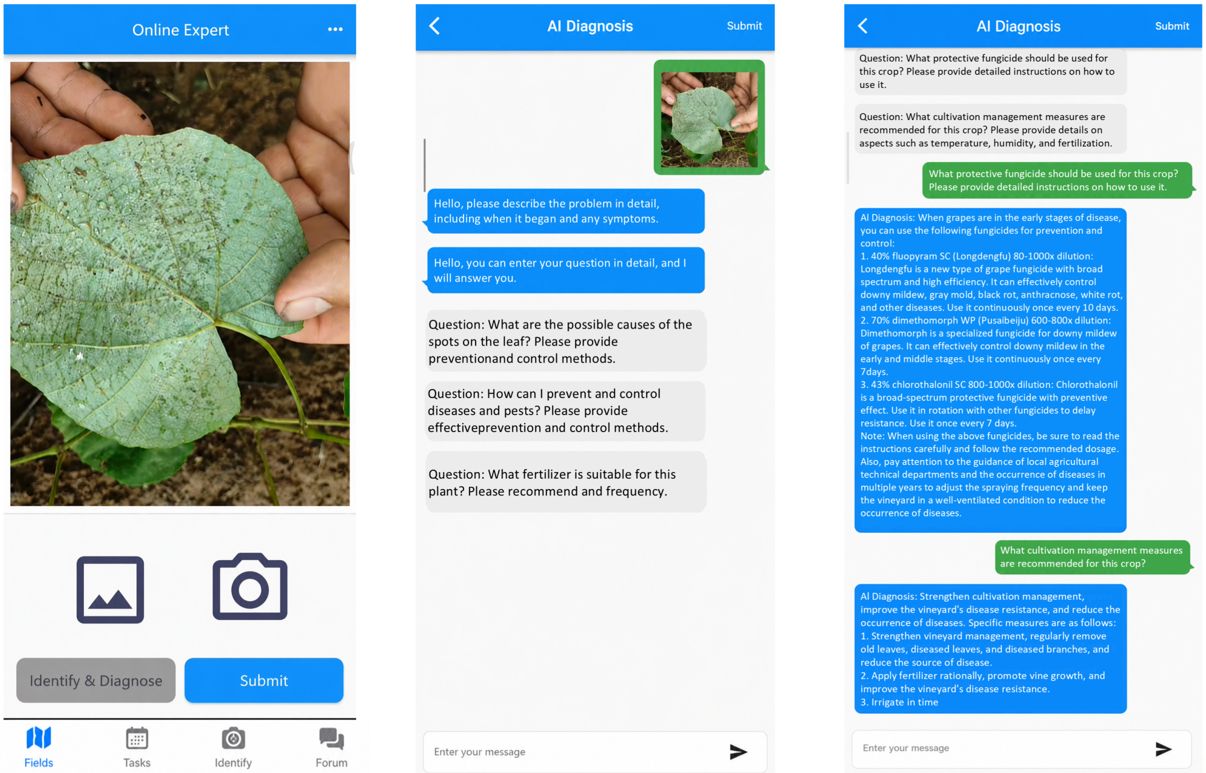
(a) **(b)** **(c)**

Figure 5. Alert and notification records for crop-season events in GrapeMaster. (a) The disease-alert dialog displays a crop-season disease-risk event. (b) The management notice dialog provides a task-related prompt. (c) The notification list archives reviewable crop-season events.



(a) (b) (c)

Figure 6. Task and product-feedback interfaces in GrapeMaster. (a) The task list records field operations and task status. (b) The product-selection page stores fungicide or pesticide entries linked to plant-protection tasks. (c) The in-tank mixing page records sprayer capacity, product dosage, plot area, water volume, and planned date.



(a) (b) (c)

Figure 7. Symptom-image and advisory evidence interfaces in GrapeMaster. (a) The image-upload page records post-symptom visual evidence. (b) The recognition-result page displays the predicted grape disease category. (c) The advisory interface stores consultation text linked to the crop-season workspace.

3.4 Record-level reconstruction of the risk-to-feedback loop

The selected loop-complete crop-season replay reconstructed the retained record chain from state updating to risk interpretation, event delivery, task execution, and treatment feedback. Under the same crop-season UUID or linked task identifiers, the replay retrieved one weather payload, one phenology payload, one disease-risk payload, 53 notification records, two completed plant-protection tasks, and four fungicide-product records (Figure 8; Table 5).

Weather, phenology, and disease-risk payloads shared the crop-season key; notifications and completed tasks were recorded under the same crop-season context; and fungicide-product records carried task identifiers.

No field-note or disease-incidence note was present under this replay anchor, although symptom-image records were present in the broader retained account set. The absence of a field-note or incidence-note record was retained as an explicit absence entry for the optional field-note branch.

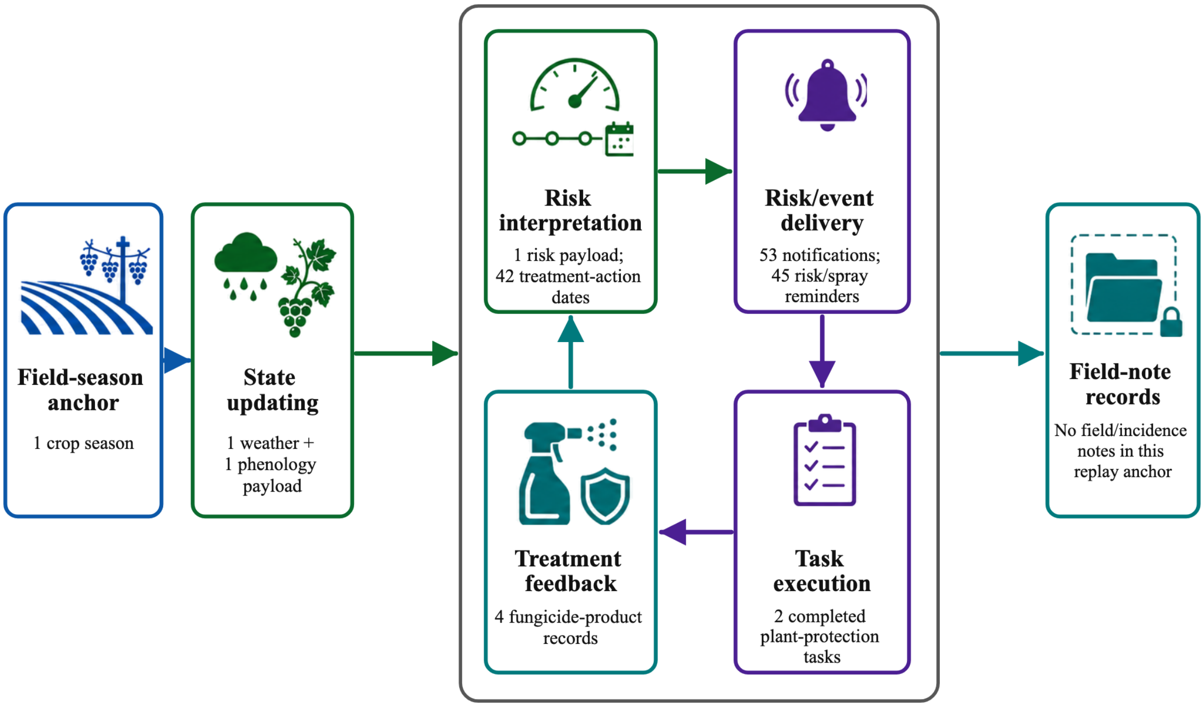


Figure 8. Record-level replay of the risk-to-feedback loop in GrapeMaster. Each node summarizes retained records under the selected crop-season management unit, and the numbers indicate record counts in this loop-complete replay. The right-hand node reports the absence of field-note or incidence-note records under this replay anchor.

Table 5. Record-level reconstruction evidence for a loop-complete crop-season replay in GrapeMaster.

Record group	Record-level evidence	Identifier or status	Result note
Replay record set	Weather, phenology, risk, notification, task, and fungicide records	One crop-season UUID	Loop-complete crop-season replay
State and risk records	Weather, phenology, and disease-risk payloads	Shared crop-season key	Shared seasonal record set
Operation records	Notifications and completed plant-protection tasks	Same crop-season context	Notification and task records
Treatment-feedback records	Fungicide-product records	Task identifiers	Product-task record set
Optional field-note branch	No field-note or incidence-note record	Explicit absence	No field/incidence note in this replay

3.5 Platform-compatible analytical module evidence

The module-level result set included downy mildew first-infection timing, field-observed grapevine growth stages, and grape disease image-recognition metrics.

The FSIM-S configuration evaluated for GrapeMaster produced a validation mean absolute error of 6.3 days and a validation root mean square error of 6.5 days for downy mildew first-infection timing (Figure 10). The image-recognition module produced class-wise F1-scores from 0.926 to 0.998 across seven grape disease categories. Downy mildew, the disease target linked to FSIM-S risk prediction, reached an F1-score of 0.986.

The phenology dataset contained 151 valid field observations from six Guangxi site-years. In grouped three-fold validation, the broad-stage configuration reached 58.0% exact major-stage agreement and 76.0% user-visible agreement after transition-window display, with a mean absolute stage distance of 0.45 major stages when the sparsely observed post-harvest class was excluded from the core metrics. In Figure 9, most non-exact cases were transition-window or adjacent-stage cases, and severe mismatches represented a small fraction of observations. Fruit development and ripening stages had lower mean stage distance than early-season stages. The wk cultivar subset contained one sparse site-year and is shown separately from the two larger cultivar subsets.

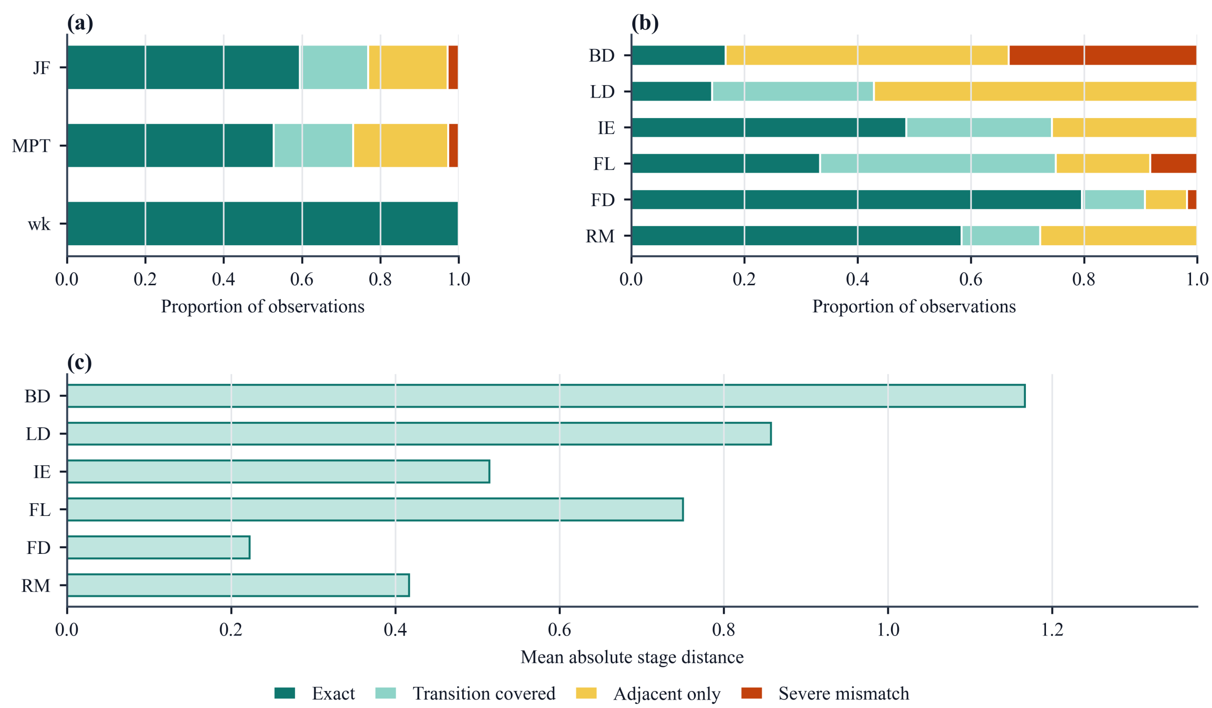


Figure 9. Validation evidence for the platform-compatible broad-stage phenology module. (a) Outcome proportions are summarized by cultivar. (b) Outcome proportions are summarized by observed major stage. (c) Mean absolute stage distance is summarized by observed major stage. BD, bud development; LD, leaf development; IE, inflorescence emergence; FL, flowering; FD, fruit development; RM, ripening of berries.

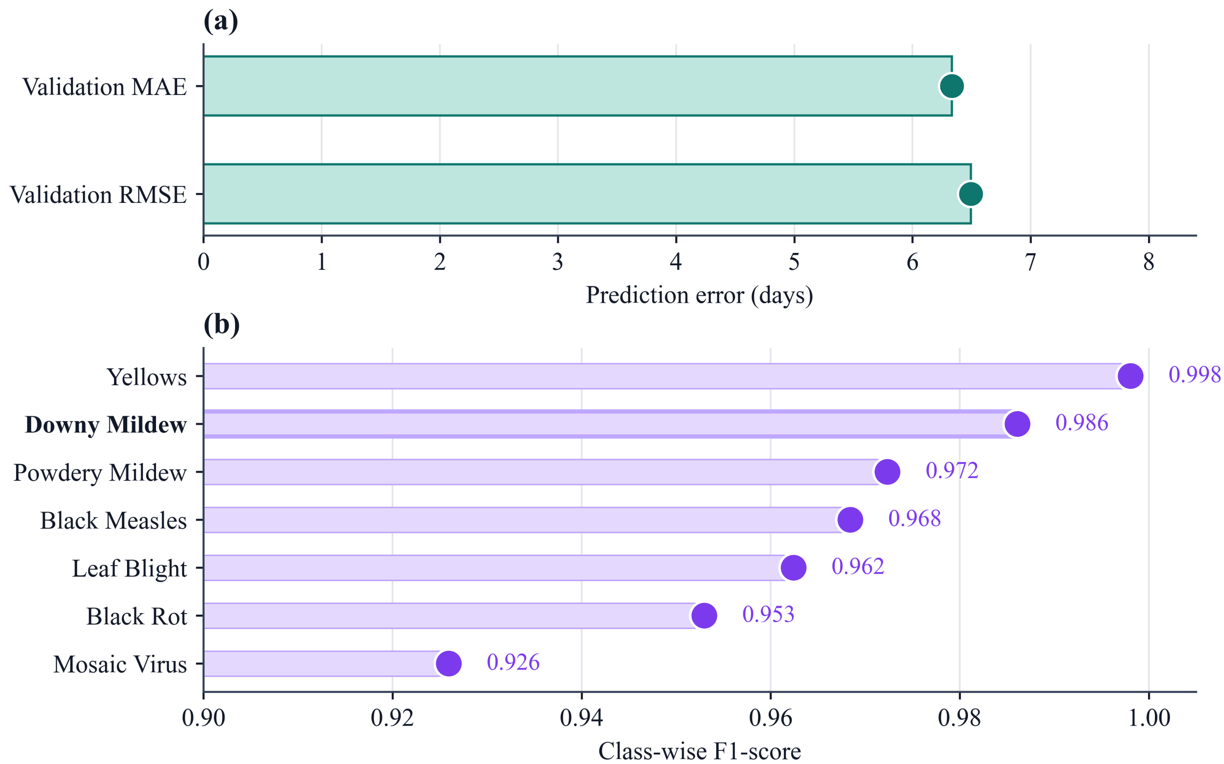


Figure 10. Module-level evidence for disease-risk and image-recognition components. (a) The FSIM-S panel reports validation error for downy mildew first-infection timing. (b) The image-recognition panel reports class-wise F1-scores for grape disease categories, with downy mildew highlighted.

4 Discussion

4.1 Crop season as the platform integration unit

In GrapeMaster, the crop season is used as the core organizational unit of the vineyard disease-management workflow, rather than as a descriptive time label or database field. This is because the vineyard field remains spatially stable, whereas weather exposure, phenological stage, inoculum pressure, protection status, observations, and management decisions are season-specific and must be reinterpreted within each crop season. This need for a season-specific management unit is consistent with previous plant disease DSS studies showing that weather-driven models and advisory systems improve decision timing only when their outputs are usable within operational contexts (Seem, 2004; Rossi et al., 2014; Bregaglio et al., 2022). It is also consistent with grapevine phenology studies showing that cultivar and regional climatic context condition crop-stage interpretation, making seasonal anchoring necessary for operational disease management rather than merely convenient for data storage (Parker et al., 2011; Wang et al., 2020).

This architecture enables cross-module retrieval through a shared seasonal unit, so related environmental, risk, operation, feedback, and evidence records can be queried without manually reconstructing links. It also makes season replay possible by keeping the risk-to-feedback sequence within a single temporal frame. For perennial grapevine production systems, repeated crop-season records from the same field can support longitudinal comparison rather than only producing episodic summaries of isolated records (Nogueira Júnior et al., 2018). These functions distinguish a crop-season-centered platform from field- or user-centered record systems in which the crop season is used mainly as a filtering label or descriptive attribute. Architecture-level linkage evidence is informative not because of record volume alone, but because retained records remained interpretable within a shared temporal and relational frame.

4.2 Traceability from risk display to field-operation feedback

GrapeMaster converts an isolated disease-risk warning into a risk-task-protection-feedback process. A warning model can identify favourable infection windows. However, a warning output does not by itself indicate that the information was reviewed, that an operation was executed, or that the resulting management context was incorporated into subsequent decision making. In vineyard disease management, the feedback step is essential because fungicide application can change the infection risk and severity of subsequent disease occurrence or renewed epidemic development. Such feedback-based adjustment can recur several times within a crop season as weather, host growth, disease pressure, and protection status change. Disease-warning studies similarly indicate that prediction value should be interpreted through action-relevant probability and decision context rather than through risk classification alone (Yuen and Hughes, 2002; Zhao et al., 2011).

The representative replay provides backend-record evidence for this traceable workflow closure (Figure 8; Table 5). Weather and phenology inputs were linked to risk display and notification, warning states were connected with plant-protection tasks, completed tasks generated fungicide-use records with product and mixing information, and these records supported subsequent protection-state interpretation and updating. Although this study did not measure agronomic efficacy, pesticide reduction, or grower adoption, it established the traceable record chain and architectural pattern required for evaluating these outcomes in future production studies.

Disease diagnosis and agronomic consultation form supplementary record branches within the same seasonal frame. They do not replace the core workflow chain, but, when present, allow observed symptoms, diagnostic outputs, consultation messages, vineyard context, and operation history to be reviewed together. This design gives growers a structured entry point for symptom-level interpretation and advisory interaction, reducing dependence on informal information sources of uneven reliability while preserving the consultation context for later review (Yao et al., 2025; Yan et al., 2025).

GrapeMaster creates a structured record basis for evaluating treatment timing, unnecessary fungicide use, disease-control outcomes, and advisory consistency in future production studies. This is particularly relevant because fungicides are a major pesticide class that directly affects production costs and raises concerns about environmental exposure (Zubrod et al., 2019). Therefore, the value of the workflow is not only that it delivers warnings, but that it converts warnings into auditable management sequences that can support outcome-oriented evaluation.

4.3 Deployment implications for vineyard digital disease management

Agricultural DSSs are increasingly becoming deployed platform services rather than isolated analytical tools. In viticulture, vite.net integrated vineyard monitoring, pest and disease models, web-based decision support, and two-way user communication to address the implementation problem of agricultural DSSs (Rossi et al., 2014), whereas MISFITS-DSS demonstrated how public research institutions and regional plant-protection services can harmonize grapevine downy mildew risk assessment across regions (Bregaglio et al., 2022). More recently, Fuxi Brain combined IoT-based perception, multimodal data fusion, and generative-model decision making for whole-cycle maize production management (Chen et al., 2026). However, their primary contributions remain concentrated in integrated monitoring and advisory delivery, regional risk-assessment harmonization, or autonomous production decision making. They do not explicitly formalize a crop-season-centered disease-management object for organizing and responding to repeated interactions among risk warnings, task execution, fungicide-protection feedback, and subsequent review across the growing season.

Against this background, the contribution of GrapeMaster is not the co-location of multiple functions in one interface, but the organization of disease-risk interpretation, crop-stage context, field task execution, treatment feedback, symptom-level diagnosis, advisory interaction, and season replay into a verifiable closed-loop workflow that uses the crop season as the shared management unit. This design is consistent with biovigilance frameworks that emphasize the connection between observations, risk interpretation, and management decisions under changing production conditions (Carisse et al., 2017).

In addition, the current deployment uses routine field records and public weather data as a lightweight baseline, while allowing orchard-scale weather stations, phenology cameras, spore samplers, or other IoT sensors to be added when finer microclimate, crop-stage, or inoculum information is needed for field-scale management (Bakir et al., 2026; Nkwocha and Chandel, 2025; Brischetto et al., 2020; Rai et al., 2026). In parallel, user-reported phenological stages, disease occurrence, treatment feedback, and field notes can support local checking and adjustment of model and advisory outputs, a requirement emphasized in DSS implementation studies where local observations and advisory workflows shape operational use (Rossi et al., 2014; Seem, 2004; González-Domínguez et al., 2023). These two extension routes allow the platform to remain lightweight at initial deployment while becoming progressively more locally adapted as sensor data and user-confirmed records accumulate; field-scale validation is still required before such adaptations are treated as operational recommendations.

4.4 Analytical modules as supporting evidence for platform readiness

Independent analytical modules provide decision context for GrapeMaster. The FSIM-S risk module supports warning of the first seasonal infection period of grapevine downy mildew, the phenology module supplies crop-stage context for interpreting susceptibility and operation timing, and image recognition adds visual diagnostic evidence when orchard disease symptoms become visible. Although the risk-prediction workflow evaluated here is centred on downy mildew, the image-recognition and LLM-assisted advisory modules provide a broader grape-disease support branch for symptom-level identification, consultation, and management discussion when other disease symptoms are observed. The LLM-assisted advisory component provides season-long consultation and management-oriented explanation within the crop-season context, but it remains distinct from validated disease models, expert diagnosis, pesticide labels, and regulated plant-protection decisions. These roles are positioned at different decision moments within the season, so the module results should be judged by their contribution to the workflow rather than as standalone algorithmic claims. The design value of this modular embedding also lies in controlled separability. The platform claim does not depend on a single fixed model: the seasonal workflow can remain stable while analytical components are independently updated, calibrated, or locally adapted. A newer image-recognition algorithm, revised phenology thresholds, or an updated disease-risk model can therefore be introduced without redefining the overall management process. This separability is important for regional deployment because humid subtropical vineyards in Guangxi differ from many temperate or Mediterranean grape-growing regions in rainfall regime, disease pressure, canopy growth, cultivation calendars, and treatment practice. These differences can shift the timing, intensity, and frequency of the same disease and limit direct transfer of warning thresholds or management rules (Seem, 2004; Yu et al., 2017; Wang et al., 2020). FSIM-S is therefore best interpreted as a locally configured first-infection component: it represents a Guangxi setting in which the first seasonal downy mildew infection can be associated with recurrent airborne asexual sporangia under humid conditions, rather than assuming a purely temperate pattern dominated by oospore-driven primary infection (Rossi et al., 2008; Caffi et al., 2009, 2011a; Brischetto et al., 2021). The same principle applies to phenology display, image recognition, and advisory support. Their value is not that they replace local agronomic judgement, but that they convert local observations, calibrated thresholds, symptom evidence, and management rules into explicit outputs that can be inspected, corrected, and combined within the crop-season workflow. When the platform is transferred to other cultivars, training systems, climatic regions, or advisory regimes, the GrapeMaster architecture supports recalibration and validation of individual modules with local evidence rather than reuse as fixed experience-based assumptions (Yan et al., 2025).

4.5 Evaluation scope, limitations, and future work

The results of this study are best interpreted as traceable workflow-closure verification and module-readiness assessment for the platform, rather than as a final agronomic-impact evaluation. Architecture-level linkage summaries, user-facing workflow records, representative replay, and module-level tests together show that the platform can generate the expected data objects around the crop-season unit, preserve linkage rules, support feedback closure, and host platform-compatible analytical modules.

Because a single grapevine crop season spans a long annual production cycle, this study did not yet obtain large-scale deployment data across repeated seasons for evaluating production outcomes such as fungicide-use efficiency, disease-control performance, and economic return. This limitation does not change the primary evidential focus of the study, which is workflow closure, traceability, and module readiness rather than final agronomic-impact assessment. Such production-level data can be accumulated in later deployment through broader regional use, repeated crop-season operation, and longitudinal user interaction. Analytical stability likewise requires longer-term monitoring across years, cultivars, and vineyard microclimates. User-level studies also remain necessary to provide longitudinal statistics on interaction efficiency, recommendation adoption, and suitability across farms of different sizes.

Methodologically, this study positions the current results as an intermediate evaluation step between architectural workflow closure of the platform and production-outcome testing. Before agronomic impact can be assessed through repeated field comparisons, a deployable digital agriculture platform should first demonstrate that its management unit, linkage rules, and workflow states can carry information from prediction to action, feedback, evidence, and review. The GrapeMaster evaluation followed this management-first sequence by defining the crop-season unit, building the workflow, embedding analytical modules, and verifying whether the expected record classes and identifiers persisted in the backend. This verification strategy clarifies what the current evidence can support and what must be tested later under production conditions.

Future work should evaluate GrapeMaster through repeated crop-season use under production conditions. Priorities include multi-region deployment, quantitative comparison with existing grower plant-protection practice, task-completion analysis, and user-centered evaluation of decision adoption. Cross-regional use will also require local checks of weather-data quality, cultivar-specific phenology calibration, disease-risk parameterization, product-rule adjustment, and advisory-content validation. The present conclusion is that GrapeMaster provides a crop-season-centered digital platform and record-verified workflow foundation for traceable vineyard disease management.

5 Conclusion

This study developed and evaluated GrapeMaster, a digital platform that uses the crop season as the core organizational unit for closed-loop and traceable vineyard disease management. The platform connects vineyard field identity, weather and phenology state, downy mildew risk interpretation, warning delivery, plant-protection tasks, treatment feedback, symptom-image records, advisory interactions, and archived management records under the crop-season management unit. The representative season replay showed that the retained backend records linked one active crop season with weather and phenology inputs, one risk-interpretation payload, 53 notifications, 45 risk or spray reminders, two completed plant-protection tasks, four fungicide-product records, and subsequent protection-state review. This indicates that GrapeMaster can convert disease-risk information into a dynamic prediction–notification–task–treatment-feedback–review sequence. Embedded module-level tests further supported the readiness of platform-compatible disease-risk, phenology, and grape disease-recognition modules for the Guangxi humid subtropical vineyard context. By turning separate analytical, advisory, and record-keeping functions into a traceable seasonal workflow, GrapeMaster extends isolated disease-management tools into a traceable, locally adaptable, and scalable management system, providing a digital and methodological foundation for more standardized vineyard-level decision processes, advisory coordination, diagnostic support, and future production-outcome evaluation.

CRediT authorship contribution statement

To be completed before submission.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The de-identified derived datasets and analysis scripts supporting the figures, tables, and summary statistics reported in this article are available from the corresponding author upon reasonable request. These materials include backend-record coverage summaries, representative crop-season replay summaries, phenology validation outputs, disease-risk timeline data, figure source data, and the code used for data summarization and visualization.

Raw operational backend exports from the deployed GrapeMaster platform are not publicly available because they contain user account records, field locations, uploaded images, messages, task records, and management information subject to privacy and confidentiality restrictions. De-identified data will be provided after confirmation to protect user privacy.

Acknowledgements

To be completed before submission.

Ethics and user data privacy statement

The backend records used in this study were analyzed in aggregated or de-identified form, and no personally identifiable user information is reported.

Supplementary material

Supplementary material associated with this article is provided as a separate file.

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